Empirical Analysis of Transformers on Graph Connectivity
Part VIII: Where Is Connectivity Information Combined?

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Abstract

Report VII opened the trunk and found a graded, split-specific completion signal that lets an unbalanced two-chain split resolve far beyond the architectural distance wall, and showed that a similarity read-out makes the model's behaviour far more robust to that signal's imperfections than the linear read-out used everywhere before it. This report does two things. First, having decided to standardize on the similarity read-out going forward, it collects the *whole* mechanistic battery — the behavioural sweep, the read-out geometry, the attention scores, the node-to-node contribution, the layer ablations, the falsification test — in one place for the first time, rather than split across a linear-read-out-first narrative. Second, it asks a new question about the mechanism itself: does the model's apparent reliance on a path's open *endpoints* reflect something the endpoints *are* (a distinguished degree-1 node), or something a path merely *has* (an end)? We test this by closing both segments of the two-chain split into cycles, removing every endpoint while keeping the split intact.

1 Introduction

1.1 What Reports I–VII established

We treat the following as established (documented, multi-seed, in Reports I–VII); this report's new work must stay consistent with them.

- **There is a sharp, architectural distance wall at 3^L** ($= 9$ at $L = 2$), which moves with depth and appears even on graph families seen $\sim 10^8$ times in training (Reports III–IV).
- **The model computes distance-bounded matrix powering, not a bounded traversal**, and a soft, learned "node budget" on dense structure is a removable data prior, not a second capacity (Report V).
- **Parallel routes and repeated local structure both help, within capacity, in data-shaped ways** rather than architecturally for free (Report VI).
- **An unbalanced two-chain split resolves far beyond the distance wall; a balanced split does not, with the same weights** (Report VI, Thread B) — and opening the trunk shows why (Report VII): on a graph with exactly two components, once one is small enough to resolve completely, the model gets an extra, graded completion signal for the *other* component, built by the same two-layer attention machinery that does ordinary beyond-capacity reach, not a separate circuit. The signal survives under *both* training distributions tested (disjoint paths, pure Erdős–Rényi) and at three canvas sizes ($n = 20, 40, 64$); a label/index shortcut is refuted throughout.
- **The read-out shape decides how much that signal can hurt, not whether the trunk has it.** The linear read-out's specific failure modes — a hard reach floor, a cut decision defended only by the sign of an unbounded logit — are properties of *that* read-out.

The similarity read-out (z_{ij} = scale$\cdot\cos(h_i, h_j)$+bias) shows the same graded, split-specific mechanistic signal (Report VII, §sec:res-similarity) but converts it into a materially more robust behaviour: at $n = 40$, reach never drops below 0.91 (against a linear floor of 0.58–0.65) and cut never drops below 0.995 (against a linear collapse to 0.15 for one training distribution and to 0.01–0.09 for another) — a global, fixed cosine margin defends the cut decision everywhere a per-pair unbounded logit could not.

1.2 The new question: where, and how, is the information combined?

Every report through Report VII reads the trunk's computation through a single lens at a time: how far it reaches, which node budget it respects, which oracle it resembles, which attention weights or read-out entries move with a split. What none of them asks directly is the more basic mechanical question: *at which point in the forward pass does the model already have the answer* — has it already decided "connected" or "not" by the end of layer 1, or only after layer 2? Does the feed-forward step (rather than attention) do the actual *combining* of "I am near this node, and that node is near the target, therefore. . . "? This report is framed around that question: not just *whether* a signal exists, but *where in the stack* it is assembled. §3 lays the groundwork by putting every existing mechanistic quantity for the (now-standard) similarity read-out in one place, including a new *causal* complement to the existing node-to-node contribution: instead of only asking how sensitive the final embedding is to a perturbation of an internal representation, it intervenes directly on the input graph (removing one edge at a time) and reads off the resulting change (§3.5). §4 tests one specific structural claim the existing evidence suggests (that a path's open ends carry special weight) by removing the ends entirely. §5 lists the concrete next steps this framing still calls for — comparing the raw per-layer embeddings $h^{(1)}, h^{(2)}$ directly (not only derived quantities) to see whether a two-component "block" structure is visible layer by layer, and inspecting the attention output added to the residual stream directly.

2 Setup

Model. The same base model as Reports III–VII: the RoBERTa-faithful standard transformer, $L = 2$, single head, $d_{\text{model}} = 512$, normalised-ReLU attention. **From this report on we standardize on the similarity read-out** (z_{ij} = scale $\cdot \cos(h_i, h_j)$ + bias, a single global scale and bias, genuinely symmetric in h_i, h_j) rather than the linear read-out used as the default through Report VII. The justification is Report VII's own finding (§1.1 above): the two-component completion signal this project has spent two reports characterising is present under *either* read-out, but the similarity read-out is categorically more forgiving of the trunk's imperfect, graded cross-component mixing — reach does not floor, cut does not collapse, and (for a model that never saw disconnected graphs in training) the mechanism-vs-behaviour dissociation is cleaner. Since the read-out is not the thing under investigation here, we fix it to the one that lets behaviour track the mechanism most faithfully. We start with the *disjoint-paths* training distribution (unions of 1–4 paths covering all n nodes, as in every two-chain result since Report VI) at $n = 40$, four seeds — the checkpoint set already used for Report VII's own similarity comparison. Other training distributions and canvas sizes are left for a later pass (§5).

Notation. We reuse Report VII's forward-pass notation without repeating the full derivation: $h_i^{(0)}$ is node i's read-in embedding (after the embedding LayerNorm); $h_i^{(1)}, h_i^{(2)}$ its embedding after each of the two transformer blocks; $S_{\ell,ij} = q_i^\top k_j/\sqrt{d_h}$ the raw attention score and $\alpha_{\ell,ij} = \frac{1}{n}\max(0, S_{\ell,ij})$ the normalised-ReLU attention weight at layer ℓ; and $C_{ik} = \|\partial h_i^{(2)}/\partial h_k^{(0)}\|_F$ the exact node-to-node contribution (real automatic differentiation through the whole forward pass — V, W_O, the residual identity, the feed-forward step, LayerNorm's true local derivative), introduced in Report VII §sec:setup to replace an attention-only rollout approximation. §5 discusses a planned revision of this last quantity.

Test graphs. Two-chain graphs partitioning all $n = 40$ nodes into two disjoint paths of size a and $n - a$, swept over $a = 1, \ldots, 20$, exactly as in Report VII.

3 The full mechanistic battery, in one setting

Report VII introduced the similarity read-out comparison as one section (§sec:res-similarity there) after five sections built around the linear read-out; consequently several quantities computed for the similarity checkpoints were never shown together, and one (the attention-score heatmap) was computed but not rendered in the report text at all. This section collects the complete battery for the model this project now treats as standard — similarity read-out, disjoint-paths training, $n = 40$ — in one place. Every quantity below reuses Report VII's already-validated computation (same scripts, same checkpoints, same four seeds); no new computation was needed for this section.

3.1 Behavioural sweep and the raw cosine logit

Table 1 and Figure 1 give the full split sweep. Exact match still falls from 1.0 to 0 as the split balances, but reach in the long component never drops below 0.91 at any split tested and recovers to 0.995 by $a = 20$ — there is no floor-and-recovery cliff down to 0.58–0.65 the way the linear read-out shows at the same canvas. Cut stays at ≥ 0.995 across the entire sweep. The exact-match values at $a = 8$–10 (0.500) hide a clean per-seed split rather than a graded degradation: two of the four seeds solve every graph exactly at each of these splits, the other two solve none — a seed-level threshold shift of a few nodes, not a within-seed partial success.

split (a, 40$-a$)	exact	reach (long)	reach (short)	cut	pred. positive rate
$(1, 39)$	1.000	1.000	–	1.000	0.951
$(4, 36)$	1.000	1.000	1.000	1.000	0.820
$(7, 33)$	1.000	1.000	1.000	1.000	0.711
$(8, 32)$	0.500	0.910	1.000	1.000	0.624
$(9, 31)$	0.500	0.917	1.000	1.000	0.603
$(10, 30)$	0.500	0.925	1.000	1.000	0.584
$(14, 26)$	0.000	0.915	1.000	0.999	0.511
$(17, 23)$	0.000	0.961	1.000	0.995	0.501
$(20, 20)$	0.000	0.995	0.995	0.996	0.499

Table 1: *Model:* base standard transformer with the *similarity* read-out ($L = 2$, single head, $d_{\text{model}} = 512$, $z_{ij} = \text{scale} \cdot \cos(h_i, h_j) + \text{bias}$), trained on a single clean distribution of disjoint paths (unions of 1–4 paths covering all n nodes), four seeds, $n = 40$. *Test set:* two-chain graphs split (a, 40$-a$), 200 independently node-relabelled graphs per split per seed. *Metrics* (mean over seeds): whole-graph exact match; pairwise reach inside the long/short path (target connected); pairwise accuracy across the two paths (target disconnected); the model's overall fraction of pairs predicted connected.

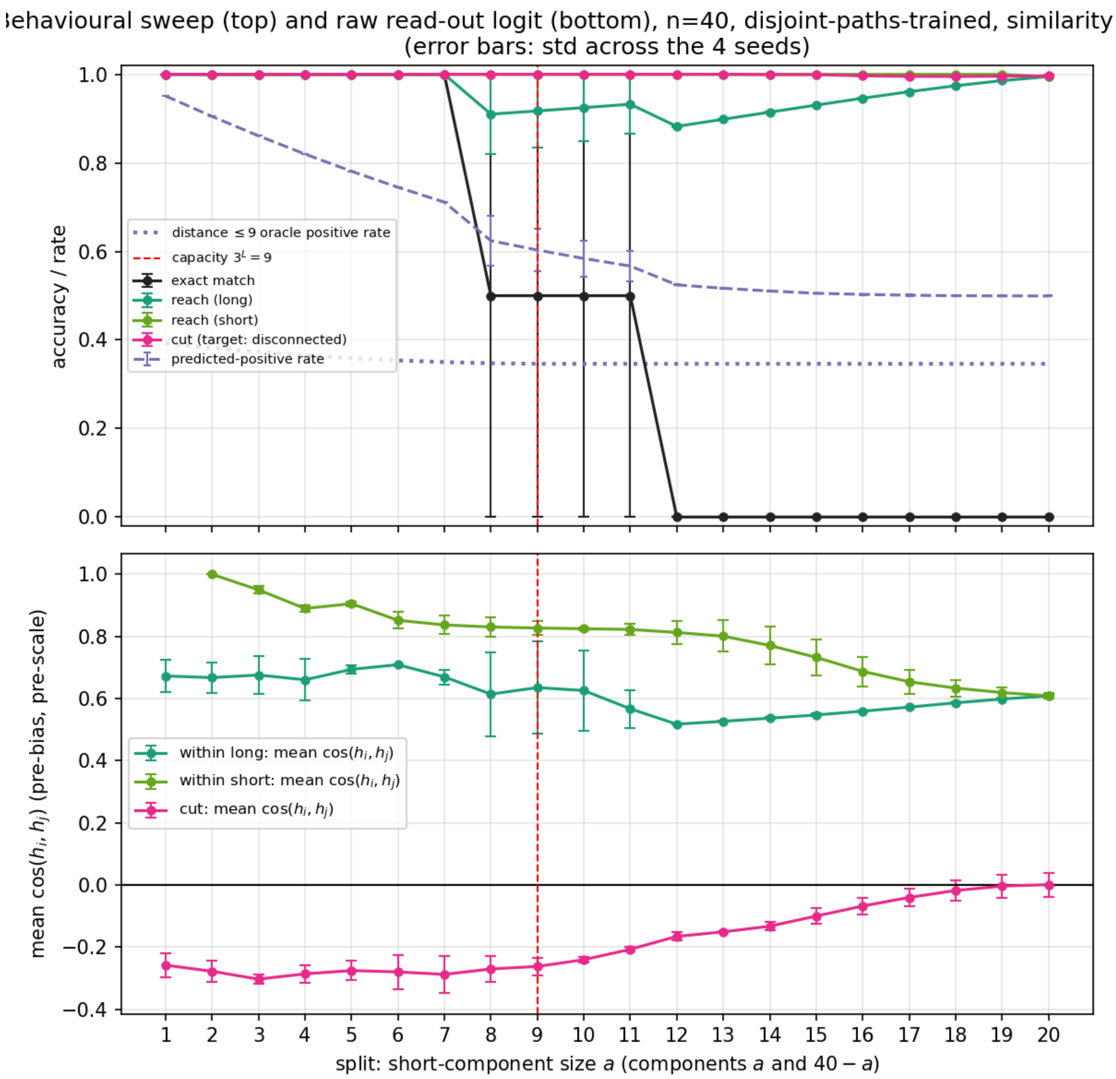

Figure 1: *Model/training/test as in Table 1*, full split sweep $a = 1, \ldots, 20$. *Top:* behavioural sweep, error bars = std across the four seeds. *Bottom:* mean raw $\cos(h_i, h_j)$ (before scale/bias) for pairs within the long path, within the short path, and across the two paths, mean over seeds, error bars = std across seeds.

3.2 Read-in geometry

There is no W_{out} for this read-out, only the read-in embedding E_{in} and two global scalars, scale and bias. Table 2 and Figure 2 give its geometry: row norms close to uniform across the 40 labels and small pairwise cosine similarity between different rows (no fixed node-index shortcut). The decision boundary is $\text{scale} \cdot \cos(h_i, h_j) + \text{bias} = 0$, i.e. $\cos(h_i, h_j) = -\text{bias}/\text{scale} \approx 0.146$ for this training distribution — a single, fixed cosine threshold every pair is measured against.

	$\|e_k\|$ (mean, std)	mean $\cos(e_k, e_l)$ $(k \neq l)$
seed 1000	1.89 ± 0.09	$+0.250$
seed 2000	1.19 ± 0.05	$+0.369$
seed 3000	2.02 ± 0.09	$+0.255$
seed 4000	1.36 ± 0.18	$+0.299$
mean scale, bias	$32.6,\ -4.77$	
implied threshold cos	≈ 0.146	

Table 2: *Model:* the four checkpoints of Table 1. *Test set:* none — static properties of the trained weights, not evaluated on any graph. *Metrics:* read-in row norms $\|e_k\|$ (mean $\pm$ std. over the 40 labels); mean pairwise cosine similarity between different rows of E_{in}; the two global read-out scalars (mean over seeds); and the cosine value at which scale $\cdot \cos +$bias $= 0$, the model's fixed connectivity threshold.

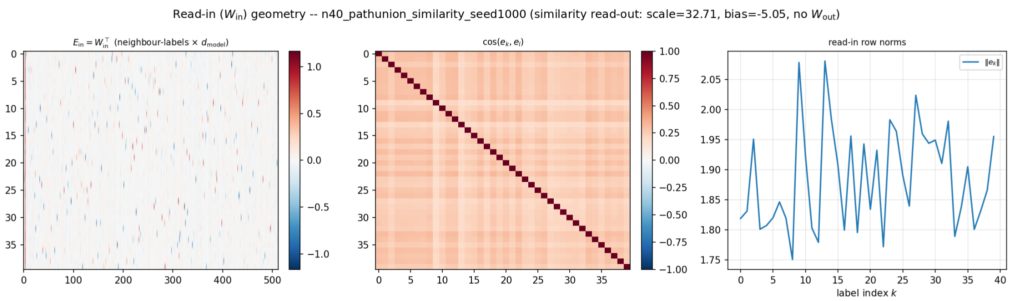

Figure 2: *Model:* the four checkpoints of Table 1. *Test set:* none — static weights, as in Table 2. *Metric:* $E_{\text{in}} = W_{\text{in}}^{\top}$ (raw entries), the cosine-similarity matrix $\cos(e_k, e_l)$, and the row norms $\|e_k\|$ by label index (one representative seed).

3.3 Attention scores and normalized-ReLU α

This figure was computed for the similarity read-out in Report VII but never shown there. Figure 3 gives the real attention-score matrix $S = QK^{\top}/\sqrt{d_h}$ and the resulting α, per layer, at $a = 4$ (solved) and $a = 20$ (failed). Layer 0 is a narrow band along the diagonal at both splits — local, structure-driven, matching every other condition in this project. Layer 1 is where the two splits differ and where a feature not visible in the linear read-out's version of this figure stands out clearly: at *both* $a = 4$ and $a = 20$, the columns nearest the two path endpoints (the first few and last few base-node indices) draw a markedly higher α than the rest of the row from almost every other node — bright vertical bands at both edges of the layer-1 panels — while the component boundary itself is visible as a break in the near-diagonal band only once the short component is a large enough fraction of the canvas (clearest at $a = 20$).

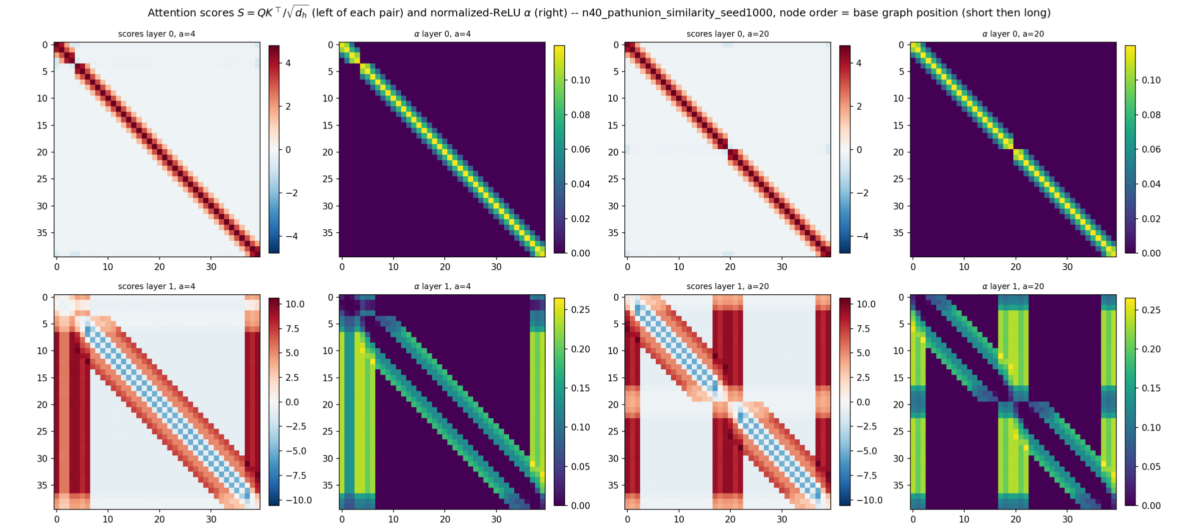

Figure 3: *Model:* one seed of the checkpoints of Table 1. *Test set:* one representative two-chain graph at $a = 4$ and one at $a = 20$, node order fixed to base graph position (short path first, then long) for readability. *Metric:* the real attention score matrix $S = QK^\top/\sqrt{d_h}$ and the resulting normalized-ReLU attention $\alpha = \mathrm{ReLU}(S)/n$, layer 0 (top) and layer 1 (bottom), averaged over 80 independent random relabellings of the same underlying graph then mapped back to base node order.

3.4 Node-to-node contribution: internal sensitivity, not yet a causal claim

Figure 4 gives the leak fraction (the share of a long-path node's exact contribution mass C_{ik} that traces back to the short component) across the full sweep, and Figure 5 the raw matrix at the same two representative splits. The leak fraction jumps from 0% at $a = 1$ to a *flat* plateau of $\approx$ 21–23% from $a = 4$ onward — neither the smooth monotone rise nor the sharp spike the linear read-out shows at the same canvas (Report VII §sec:res-attention). The matrix itself shows a genuine, non-trivial band of cross-component mixing at *both* splits, even though behaviour is nearly perfect at $a = 4$: a real dissociation between mechanism and behaviour — the trunk mixes across the component boundary by a similar amount whether or not the split is solved, and it is the read-out, not the mixing, that decides whether that costs anything (§1.1). One caveat about what this quantity actually shows, worth stating plainly rather than only in a footnote: C_{ik} is the sensitivity of $h_i^{(2)}$ to an *arbitrary* perturbation of $h_k^{(0)}$, and $h_k^{(0)}$ is itself an internal representation built from node k's *whole* neighbourhood (through the read-in and a LayerNorm), not a quantity tied only to node k's own identity. A large C_{ik} is therefore good evidence of internal token-to-token *sensitivity*, but not by itself evidence that node k's actual incident edges have a *causal* effect on node i — that question is taken up directly in §3.5, which intervenes on the input graph itself rather than on an internal embedding.

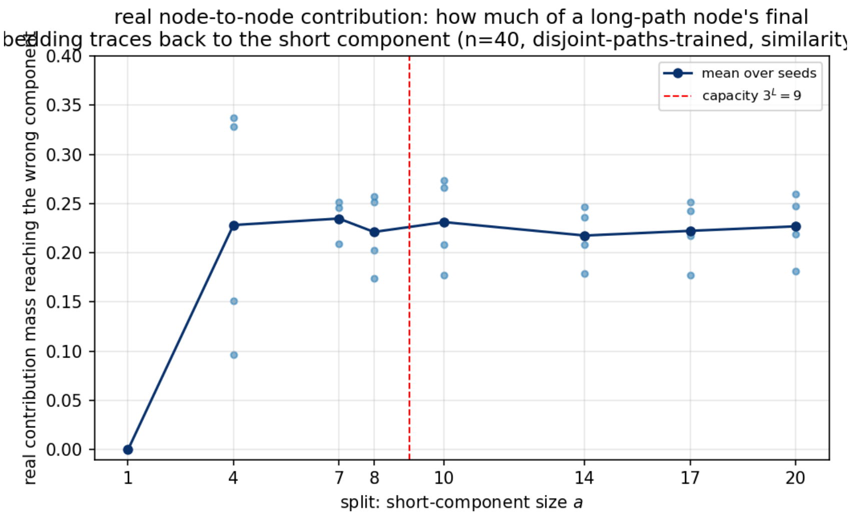

Figure 4: *Model/training as in Table 1. Test set:* two-chain graphs at the splits shown, 64 independently node-relabelled graphs per seed. *Metric:* for each long-component node, the fraction of its exact node-to-node contribution mass $C_{ik} = \|\partial h_i^{(2)}/\partial h_k^{(0)}\|_F$ (§2) that traces back to the short component rather than its own long component, mean over long-component nodes; small dots are individual seeds, the line is the mean over seeds.

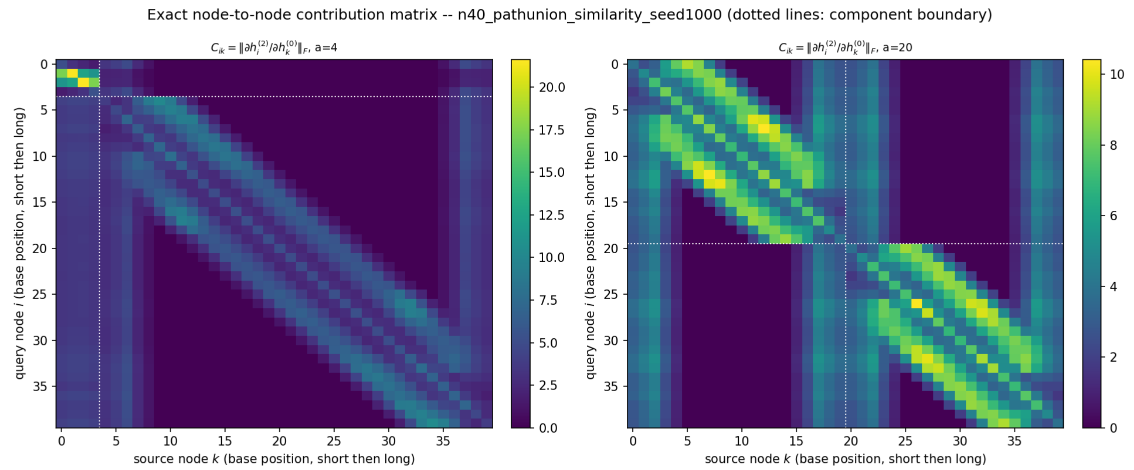

Figure 5: *Model:* one seed of the checkpoints of Table 1 (representative; the pattern is consistent across seeds). *Test set:* two-chain graphs at the split shown ($a = 4$ left, $a = 20$ right), 64 independently node-relabelled graphs, C_{ik} averaged over them. *Metric:* the raw exact node-to-node contribution matrix $C_{ik} = \|\partial h_i^{(2)}/\partial h_k^{(0)}\|_F$, rows/columns ordered by base position (short component first, then long). Dotted white lines mark the component boundary.

3.5 A causal complement: edge-ablation contribution

Why a second measure. §3.4 measures sensitivity to a perturbation of an already-processed embedding. To make a *causal* statement about the short component's edges — do they actually drive the long component's beyond-capacity predictions, or does the trunk merely correlate the two — the natural experiment intervenes on the input graph directly: remove one edge, re-run the forward pass, and measure what changes.

Construction. For every edge $e = \{u, v\}$ of a two-chain graph (both directions removed from A' at once, since the model is trained on symmetric inputs; self-loops on the diagonal are left untouched), we compare a forward pass on the intact graph against one with e removed, and read three quantities off node i: the change in its final embedding $\|h_i^{(2)}(A'\backslash e) - h_i^{(2)}(A')\|_2$; the mean absolute change over its whole logit row; and the same logit change restricted to the pairs (i, j) that matter most for the beyond-capacity puzzle — both i, j inside the long component, at true graph distance > 9. Averaging (never summing, so that a path endpoint's degree of 1 is not confounded with an internal node's degree of 2) over every edge touching a given node k turns each of the three quantities into a 40×40 matrix, column k = which node's incident edges were removed, row i = whose state we observe — the same shape as Figure 5, but built from real interventions on the graph rather than a derivative of an internal embedding. One honest caveat carried through the design rather than left implicit: on a path or a cycle *every* edge is a bridge for at least one pair, so removing it also changes that pair's true connectivity target — this measures the effect of a real structural change, not an independent, additive contribution the way the phrase "edge contribution" might suggest.

[Data pending — the computation is implemented and smoke-tested (a forward-pass-only intervention, no backward pass, cheaper than §3.4's Jacobian) but not yet run on the real checkpoints at the time of writing. Replace this paragraph with the three heatmaps at $a = 4$ *and* $a = 20$*, the edge/logit/far leak-fraction curve across the full split sweep, and a comparison against §3.4's reading once the job returns. The prediction to check: at* $a = 4$ *the short component's edges should show a real effect on the long component's beyond-capacity logits (a high* C^{far} *leak); at* $a = 20$ *that effect should be small and confined near the diagonal.]*

3.6 Layer ablation

Table 3 and Figure 6 give the ablation sweep. The bypass condition (both transformer blocks skipped, read-out applied directly to the read-in embedding) already gives cut 0.95–0.995 across the whole sweep — higher than several of the attention-ablated conditions — while its reach is low throughout (0.10–0.20): most of what makes cut work is already present in the read-in embeddings' cosine geometry before any attention runs, and the trunk's main job for this read-out is building reach, not cut. Zeroing attention layer 1 knocks reach down only mildly (0.82–0.89) while zeroing layer 0 leaves reach essentially untouched (0.987–0.992) but collapses cut to 0.01–0.04 once the split is no longer tiny — the reverse of the dependency the linear read-out showed, where cut depended on layer-1 feed-forward specifically.

condition	a=1	a=4	a=7	a=8	a=10	a=14	a=17	a=20
reach (long component)								
baseline	1.000	1.000	1.000	0.911	0.925	0.915	0.961	0.995
zero attn 0	0.992	0.992	0.992	0.992	0.991	0.990	0.988	0.987
zero attn 1	0.885	0.876	0.868	0.865	0.859	0.844	0.831	0.820
zero mlp 0	0.959	0.954	0.953	0.959	0.949	0.942	0.937	0.923
zero mlp 1	0.794	0.944	0.913	0.886	0.902	0.921	0.961	0.991
bypass	0.102	0.110	0.120	0.124	0.132	0.152	0.172	0.197
cut (target: disconnected)								
baseline	1.000	1.000	1.000	1.000	1.000	1.000	0.996	0.997
zero attn 0	0.997	0.035	0.025	0.024	0.019	0.015	0.013	0.014
zero attn 1	0.752	0.903	0.382	0.343	0.293	0.241	0.225	0.219
zero mlp 0	0.917	0.047	0.136	0.113	0.100	0.082	0.074	0.074
zero mlp 1	0.807	0.226	0.335	0.477	0.518	0.686	0.739	0.760
bypass	0.949	0.986	0.991	0.992	0.993	0.994	0.995	0.995

Table 3: *Model/training as in Table 1*, four seeds. *Test set:* two-chain graphs at the splits shown, 150 graphs per seed, independently node-relabelled. *Metric:* reach (long component, top block) and cut (bottom block), mean over seeds, under six forward-pass conditions — unmodified (baseline); the attention or feed-forward branch zeroed at layer 0 or 1; and both transformer blocks skipped entirely (bypass, read-out applied directly to the read-in embedding).

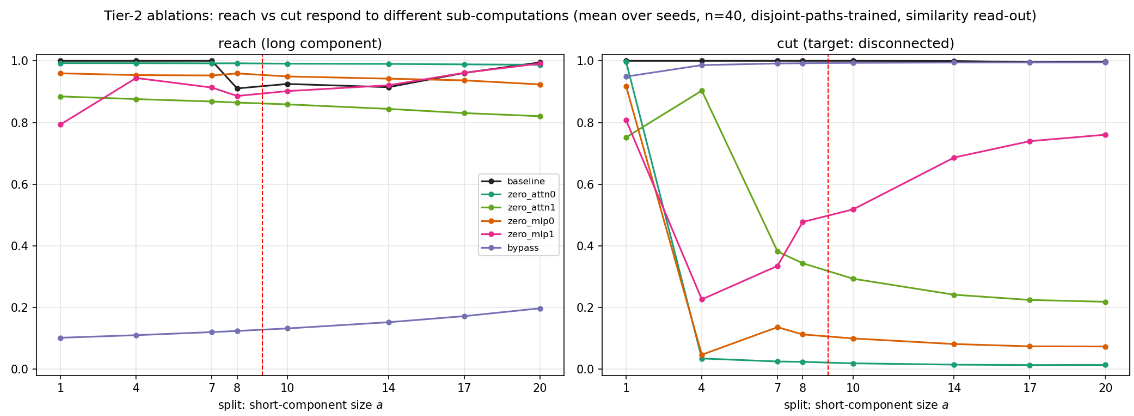

Figure 6: Table 3 as curves. *Model/training/test as in Table 3*, mean over four seeds. Red dashed line: capacity $3^L = 9$.

3.7 Falsification test: three components

Table 4 repeats Report VII's falsification construction: a small, cleanly-resolvable path plus two large paths that are *not* connected to each other. If the model implemented a crude "not the small one therefore connected" rule, it would wrongly merge the two large components. It does not, at any small-component size tested: $\mathrm{cut}(\mathrm{L}_1, \mathrm{L}_2) \geq 0.988$ throughout, reaching 1.000 for several cells, with reach inside each large component close to perfect once the small component is not itself a single node.

small	reach L_1/L_2	cut(S,L)	cut(L_1,L_2)
1	0.998	1.000	0.998
2	1.000	1.000	0.988
4	1.000	1.000	0.999
7	1.000	1.000	0.999
8	1.000	1.000	1.000
10	1.000	1.000	1.000

Table 4: *Model/training as in Table 1. Test set:* three disjoint paths partitioning all 40 nodes — one small path (size shown) and two large paths, the two large paths *not* connected to each other; 300 graphs per seed, independently node-relabelled. *Metrics* (mean over four seeds): pairwise reach inside the two large components (mean of L_1, L_2, target connected); pairwise accuracy between the small path and either large path (target disconnected, mean); pairwise accuracy between the two large paths (target disconnected — the decisive column).

4 Testing the path-endpoint hypothesis: closing the chains into cycles

Where this comes from. Two independent pieces of evidence, both in §3 and in Report VII, single out a path's open *endpoints* (its two degree-1 nodes) as structurally unusual. Under the linear read-out (Report VII §sec:res-heatmaps), the far endpoint of each path segment is an outsized *source* of node-to-node contribution — 3–4× the per-node average — without being an unusual *receiver*, read there as a "this path is closed" signal broadcast outward. Under the similarity read-out (§3.3 above), the columns nearest each path's two endpoints draw markedly more layer-1 attention than the rest of the graph, at every split tested. Both readings point the same way, but neither rules out the alternative: that these effects are an artefact of *how the test graphs are built* (a path always has exactly two special nodes) rather than something the model's completion mechanism actually depends on.

The test. We remove every endpoint while changing nothing else: close each of the two path segments into a **cycle**, keeping the same split $(a, n-a)$, the same total node count, and the same disjoint-two-component structure — only now neither component has a degree-1 node. A cycle needs at least three nodes, so the sweep starts at $a = 3$ rather than $a = 1$. We feed this construction to the *same* checkpoints as §3 (disjoint-paths-trained, similarity read-out, $n = 40$) — eval-only, no retraining — and read the identical battery: the behavioural sweep, the exact-Jacobian contribution leak fraction of §3.4, and the causal edge-ablation contribution of §3.5, all with no change to either measure's definition — only the underlying graph topology changes. One adjustment the cycle geometry itself forces: a k-cycle's diameter is $\lfloor k/2 \rfloor$, roughly half a k-node path's, so the beyond-capacity pairs the far-logit variant of §3.5 looks at only exist once the long *cycle* segment is long enough (around 19 nodes, versus around 10 for a path of the same role).

Two readings, stated before looking at the data. If the model solves the cycle sweep about as well as the chain sweep (same split threshold, similar reach/cut pattern), the endpoints were *incidental*: whatever the completion mechanism actually tracks does not depend on a degree-1 node existing, and the source/high-attention effect at the endpoints was a by-product of them being the two nodes farthest from every other node on their segment, not a special role tied to being an open end. If instead the model's performance degrades sharply or the split threshold moves, the endpoints are *load-bearing*: removing them removes something the mechanism relies on to decide a segment is fully resolved.

[Data pending — the eval is prepared (eval-only on the existing similarity checkpoints, no training) but not yet run at the time of writing. Replace this paragraph with the real sweep table, the leak-fraction comparison against Figure 4, the edge-ablation comparison against §3.5, and the verdict once the job returns.]

5 Outlook

The causal edge-ablation contribution of §3.5 answers the concern that motivated it (§3.4): it is implemented and its design is fixed, only the data collection is still pending. Beyond it, three concrete next steps follow from the framing of §1.2, none implemented yet.

- **A chord-addition control for the edge-ablation measure.** On a path or a cycle every edge is a bridge, so removing one always changes the true connectivity target for at least one pair (§3.5) — the measured effect is real but confounded with that target change. Repeating the same intervention with an edge *added* between two nodes already in the same component (a chord) leaves the true target unchanged and would isolate the effect of a structural change from the effect of a target change.
- **Plot the per-layer embeddings themselves, not only derived quantities.** Every mechanistic result so far (attention, contribution, read-out logits) is a quantity *derived from* $h^{(1)}$ and $h^{(2)}$. Plotting the embeddings directly (e.g. a low-dimensional projection, coloured by which component a node belongs to) at each layer would show directly whether a two-component "block" structure is visible in the representation itself, and at which layer it first appears.
- **Inspect where the information is combined.** The working hypothesis is that the trunk first establishes, per node, something like "which other nodes are near me" and only later *combines* that with the target's identity to decide connectedness — and that this combination step is a natural place to look for in the feed-forward sub-layer rather than attention. Plotting the attention output added to the residual stream directly (rather than only the attention weights α) at each layer is the first concrete probe of this, to see whether the two branches are already combined before the feed-forward step or only after it.

Beyond these three, the battery of §3 and the cycle test of §4 are both, for now, run only on the disjoint-paths training distribution at $n = 40$. Extending either to the pure-Erdős–Rényi distribution and to $n = 20/64$ (as Report VII did for the linear read-out) is the natural next pass once the questions above are settled.